

MSBRDM: Human Robot Collaboration

Richard Fabian Willke

*Electrical Engineering and Information Technology
Technical University Munich
Munich, Germany
go34qil@mytum.de*

Rohan Singh

*Robotics, Cognition and Intelligence
Technical University Munich
Munich, Germany
rohan.singh@tum.de*

Mohamed Altorky

*Robotics, Cognition and Intelligence
Technical University Munich
Munich, Germany
mohamed.altorky@tum.de*

Human-Robot Collaboration is one of the most prominent aspects of robotics research in recent years. The common goal of this field is to design controls for machines that enable them to safely and reliably cooperate with a human operator toward a shared goal. To achieve this, several system architectures and controllers have been implemented, all aiming to assist the human operator in their tasks. This report presents an approach on a UR10 robot manipulator with a lacquey gripper that uses a mixture of error space joint control, error space Cartesian control, and admittance control, where the admittance controller generates references for the error space Cartesian controller once activated by the human operator.

I. INTRODUCTION

The key to successful human-robot collaboration is a controller architecture that can reliably interpret the human's intent based on the wrench applied by the human and then act accordingly. This report specifically focuses on manipulation tasks using robot arms. In this specific use case, the robot arm is used to ease the load of a cargo stored in a crate. To reliably assist the human with the weight of the load, the robot arm needs to be able to reliably estimate the additional weight on the end-effector and correctly read and interpret the input from the user. The goal of this project is to guide the motion of the robot in Cartesian space using the force applied by the human. For this project, a UR10 robot arm manipulator, equipped with a lacquey gripper as the end-effector, is used. Positioned between the gripper base and the robot arm's end is a force-torque sensor. This sensor is used to measure the 6D wrench applied by the user onto the end-effector. The direction and magnitude of the force are then translated to a desired position and orientation of the end-effector in Cartesian space using admittance control.

II. METHODOLOGY

The core of this project is to control the UR10 robot arm such that it compliantly follows the external wrench applied by the user on its end-effector. To achieve the desired results, we design several controllers, each suited for different domains of

operation. To address modeling uncertainties, sensor noise, and dynamic variations, every controller employed in the system is wrapped by an **adaptive controller**. This adaptive layer enables the system to adjust its control parameters in real-time, ensuring robust performance regardless of the control domain. In this specific case, we designed adaptive controllers in the error space, both for Cartesian and joint position control. The Cartesian controller is enhanced by an admittance controller for when the human interacts with the robot arm. The admittance controller is driven by the magnitude and direction of the external wrench applied by the human onto the gripper. To increase the reliability of the sensor readings, a Butterworth filter is implemented. A diagram of the system is displayed in Figure 1.

The entire system can be described using the state machine depicted in Figure 2 and the RQT node graph shown in Figure 3. The robot utilizes the individual controllers sequentially. To access the admittance controller, the human operator must call a ROS service on the PC that is connected to the robot

1. Adaptive Control Strategy

The adaptive controller monitors the error signals generated by the underlying controllers and updates an adaptive parameter vector Θ to compensate for uncertainties. In particular, a regressor matrix Y_r is computed based on the current joint states and the generated reference signals. The adaptation law is defined as

$$\Theta \leftarrow \Theta - \eta Y_r^\top S_q \quad (1)$$

where:

- Y_r is the regressor matrix,
- S_q is the joint-space error (for example, $S_q = \dot{\theta} - Q_{pr}$ where Q_{pr} is the reference joint velocity computed from the lower-level controllers),
- η is the learning rate.

The final control torque is then computed by merging a derivative feedback term with the adaptive compensation:

$$\tau = -K_d S_q + Y_r \Theta \quad (2)$$

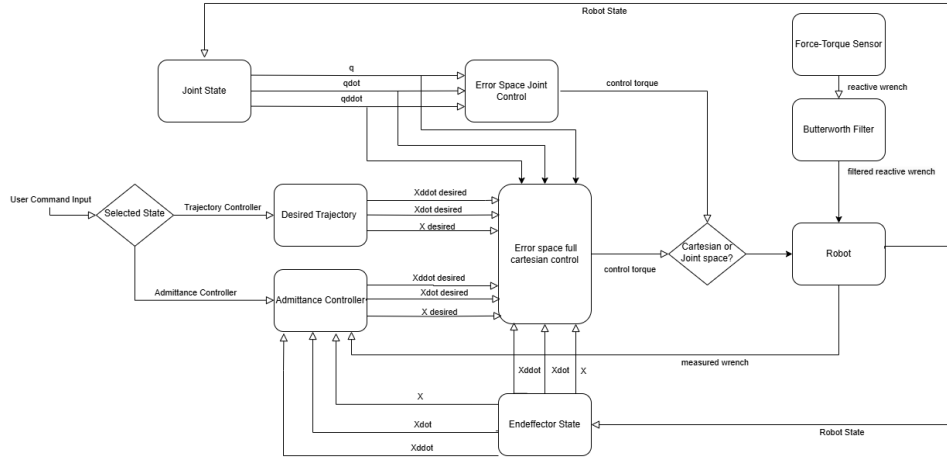


Fig. 1: Overall control structure of the system

where K_d is the derivative gain.

This adaptive control law is uniformly applied across all controllers (Cartesian, joint, and admittance). By doing so, the system is capable of dynamically compensating for uncertainties and delivering robust performance throughout the diverse control strategies used in the project.

Throughout the project, different controllers—such as the Cartesian space controller, joint space controller, and the admittance controller—compute a joint space error, denoted as S_q . Specifically, S_q represents the difference between the actual joint velocity $\dot{\theta}$ and the computed reference joint velocity Q_{pr} , mathematically expressed as:

$$S_q = \dot{\theta} - Q_{pr} \quad (3)$$

This joint space error is then passed to the adaptive controller to update the controller parameters in real time, ensuring robust performance across the varying controller domains.

2. Error Space Controllers

2a. Error space joint controller: Joint error is defined as:

$$\Delta\theta = \theta - \theta_d, \quad \Delta\dot{\theta} = \dot{\theta} - \dot{\theta}_d \quad (4)$$

A PID controller in joint space computes the reference joint velocity:

$$Q_{pr} = \dot{\theta}_d - K_p \Delta\theta - K_i \int \Delta\theta dt \quad (5)$$

and the reference joint acceleration:

$$Q_{ppr} = \dot{\theta}_d - K_p (\dot{\theta} - Q_{pr}) \quad (6)$$

Defining the joint error as

$$S_q = \dot{\theta} - Q_{pr} \quad (7)$$

Q_{pr} and Q_{ppr} are used to calculate the regressor matrix Y_r and the resultant S_q will be used in equation 2 to compute the necessary joint torques.

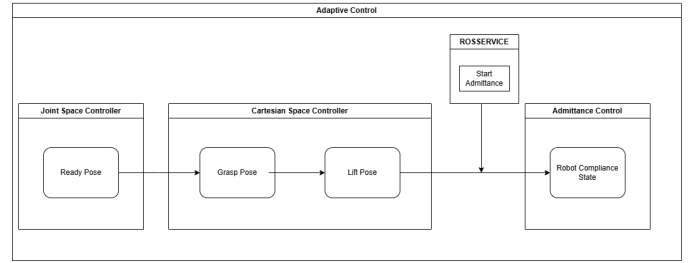


Fig. 2: System state machine

2b. Error space full cartesian controller: In an error space cartesian controller, we first calculate the references and then map them to joint space error using the inverse damped jacobian, where cartesian error is defined as:

$$\Delta X_{lin} = X_{lin} - X_{d_lin} \quad (8)$$

Further the linear references as calculated as follows:

$$\dot{X}_{ref_lin} = \dot{X}_{d_lin} - K_p \Delta X_{lin} - K_i \int \Delta X_{lin} \quad (9)$$

$$\ddot{X}_{ref_lin} = \ddot{X}_{d_lin} - K_p (\dot{X}_{lin} - \dot{X}_{d_lin}) - K_i \int \Delta X_{lin} \quad (10)$$

Given the current end effector orientation represented as a quaternion $q_{current}$ and the desired orientation $q_{desired}$, the angular error is computed as follows:

1) Compute the relative quaternion error:

$$q_{err} = q_{desired} \cdot q_{current}^{-1} \quad (11)$$

2) Normalize the quaternion:

$$q_{err} \leftarrow \frac{q_{err}}{\|q_{err}\|} \quad (12)$$

3) Convert the quaternion error to an axis-angle representation:

$$\text{AngleAxis}(q_{err}) = (\theta, \mathbf{v}) \quad (13)$$

where θ is the rotation magnitude (angle) and $\mathbf{v}$ is the unit rotation axis.

4) Compute the angular error:

$$\omega = \theta \mathbf{v} \quad (14)$$

The reference velocity vector $\dot{X}_{\text{ref}}$ is composed of the linear velocity and angular error:

$$\dot{X}_{\text{ref}} = \begin{bmatrix} \dot{X}_{\text{ref_lin}} \\ \omega \end{bmatrix} \quad (15)$$

where: $\dot{X}_{\text{ref}} \in \mathbb{R}^3$ is the linear velocity, $\omega \in \mathbb{R}^3$ is the angular error. The reference acceleration vector $\ddot{X}_{\text{ref}}$ is given by:

$$\ddot{X}_{\text{ref}} = \begin{bmatrix} \ddot{X}_{\text{d_lin}} \\ \mathbf{0}_{3 \times 1} \end{bmatrix} \quad (16)$$

where: $\ddot{X}_{\text{d_lin}}$ represents the desired linear acceleration components. The last three elements are set to zero, assuming no angular acceleration is considered. Now the references are mapped to joint error space error using the inverse damped jacobian:

$$Q_{pr} = J^{-1} \dot{X}_{ref} \quad (17)$$

$$Q_{ppr} = J^{-1}(\ddot{X}_{ref} - (\dot{J}\dot{\theta})) \quad (18)$$

$$S_q = \dot{\theta} - Q_{pr} \quad (19)$$

Now we follow a similar approach as used in the error space joint controller. Q_{pr} and Q_{ppr} are used to calculate the regressor matrix Y_r and the resultant S_q will be used in equation 2 to compute the necessary joint torques.

3. Butterworth Low-Pass Filter

In our project, a Butterworth low-pass filter was applied to the data obtained from the force-torque sensor to reduce high-frequency noise and smooth the measurements. Force-torque sensors typically capture signals that include noise from sources such as electrical interference or mechanical vibrations, which can lead to aberrant behavior.

A Butterworth low-pass filter was chosen to filter the output of the force-torque sensor due to its characteristic of providing a maximally flat response in the passband, meaning it does not introduce any ripples or distortion to the desired low-frequency signal. The filter allows signals with frequencies lower than the cutoff frequency, ω_c , to pass through with minimal attenuation, while significantly attenuating higher-frequency components beyond the cutoff.

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}}} \quad (20)$$

where:

- ω_c is the cutoff frequency.

- N is the order of the filter.
- ω is the frequency variable.

where the poles are given by:

$$p_k = \omega_c e^{j \frac{(2k+N-1)\pi}{2N}}, \quad k = 0, 1, 2, \dots, N-1 \quad (21)$$

For a second-order ($N = 2$) Butterworth filter:

$$H(s) = \frac{1}{1 + \sqrt{2} \frac{s}{\omega_c} + \left(\frac{s}{\omega_c}\right)^2} \quad (22)$$

The roll-off rate of the Butterworth filter is:

$$\text{Roll-off} = -20N \text{ dB/decade} \quad (23)$$

For a second-order Butterworth filter ($N = 2$) as the frequency approaches zero ($\omega \rightarrow 0$):

$$|H(j\omega)| = \frac{1}{\sqrt{1+0}} = 1 \quad (24)$$

This indicates no attenuation in the passband.

Further as the frequency approaches infinity ($\omega \rightarrow \infty$):

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\infty}{\omega_c}\right)^2}} \rightarrow 0 \quad (25)$$

This means the signal is strongly attenuated in the stopband.

For our task ω_c was set to 10 Hz and sampling frequency to 400 Hz and the comparison of the raw sensor data to the filtered data is shown in figure 4. The Butterworth filter significantly improved the signal quality by attenuating high-frequency noise, thus providing cleaner and more accurate force and torque measurements, which allowed more reliable performance in admittance control task.

4. Admittance Controller

Admittance control is a control strategy widely used in robotics to achieve compliant interaction with the environment. Unlike impedance control—which regulates force based on position error, admittance control maps external forces directly into motion commands. In this approach, the measured external wrench is interpreted as an input to a dynamic system defined by virtual inertia, damping and stiffness, and it computes a corresponding desired acceleration. This computed acceleration is then integrated to generate smooth end-effector trajectories, ensuring safe and adaptive behavior even during unexpected contact or disturbance.

We implement an admittance controller for a 6-DOF robot that seamlessly converts external force measurements into compliant motion. The main steps are:



Fig. 3: RQT node graph

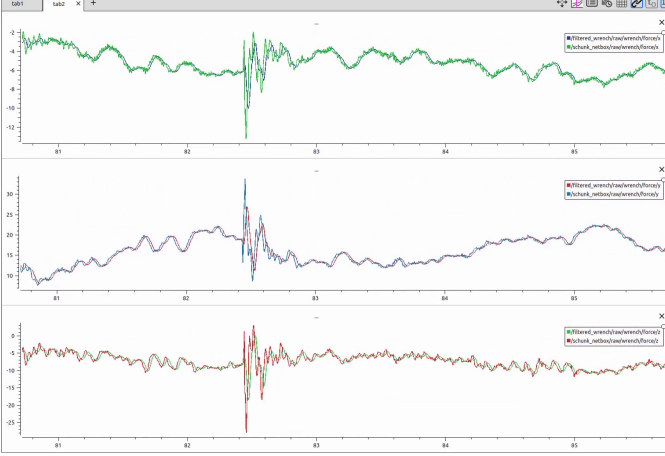


Fig. 4: Results of the Butterworth low pass filter on the force-torque sensor data.

When the external force exceeds a threshold, the desired Cartesian acceleration is computed as:

$$\ddot{X}_d = \Lambda_{adm}^{-1} (W - K_{damp} \dot{X}_{ee} - K_{stiff} \Delta X_{ee}) \quad (26)$$

where:

- W is the measured external wrench (force/torque),
- Λ_{adm} is the admittance matrix,
- $\dot{X}_{ee}$ is the end-effector velocity.
- X_{ee} is the end-effector position.

This formulation allows the robot to react appropriately to varied external forces, making admittance control particularly suitable for human-robot collaboration and environments with uncertain contact conditions. The updated linear cartesian state is obtained by integrating the acceleration:

$$\dot{X}_{d_lin} = \dot{X}_{ee_lin} + \ddot{X}_{d_lin} \Delta t \quad (27)$$

$$X_{d_lin} = X_{ee_lin} + \dot{X}_{d_lin} \Delta t \quad (28)$$

For the angular part, the velocity update follows:

$$\omega_d = \omega + \dot{\omega}_d \cdot dt \quad (29)$$

where ω is the current end effector linear velocity. The quaternion update is computed using the exponential map:

$$q_{exp} = \exp_{map}(\omega_d, dt) \quad (30)$$

$$q_{k+1} = q_{exp} \cdot q_k \quad (31)$$

where q_k is the current quaternion orientation of the end effector. Finally, the desired orientation in Roll-Pitch-Yaw (RPY) angles is:

$$rpy = \text{RPY}(q_{k+1}) \quad (32)$$

The desired velocity $\dot{X}_d$ is composed of the linear velocity and angular error:

$$\dot{X}_d = \begin{bmatrix} \dot{X}_{d_lin} \\ \omega_d \end{bmatrix} \quad (33)$$

The desired position vector X_d is given by:

$$X_d = \begin{bmatrix} X_{d_lin} \\ rpy \end{bmatrix} \quad (34)$$

And finally the desired position, velocity and accelerations are fed to the error space full cartesian controller to make the robot follow the external wrench applied by the user.

III. EVALUATION

Testing of the individual modules concluded with successful performance of the Butterworth filter and the adaptive controllers in both joint space and Cartesian space. As a result, the robot was able to precisely navigate through the environment, achieving specific goal positions in both joint space and Cartesian space, including orientation. In simulation, the robot arm behaved as expected when testing the admittance controller. The admittance controller was tested using artificial wrenches applied to the end-effector. The success criterion was that the end-effector moved correctly in the desired direction with acceptable velocity and acceleration. The simulations confirmed that the controller worked as intended. The tracking results from the real robot are shown in Figure 5.

We further provide video samples and code¹ showing the result of our implementation on the UR10 robot.

CONCLUSION

Tests and the final demonstration yielded highly satisfactory results for all controller components. Parameter tuning was conducted to a degree that enabled the robot to successfully track the desired position and velocity in an impressive manner. However, we acknowledge that the tracking of orientation still requires refinement. For future work, we aim to enhance the performance of orientation tracking and further fine-tune the control parameters. Furthermore, to ensure system safety, we will implement a safety stop mechanism when the manipulability index falls below a certain threshold, at which point the robot will be reset to a hold position after which the user can then call the service again to activate the admittance control.

¹<https://syncandshare.lrz.de/getlink/fi4PmNAUx3eBaDaH6KrKkA/>

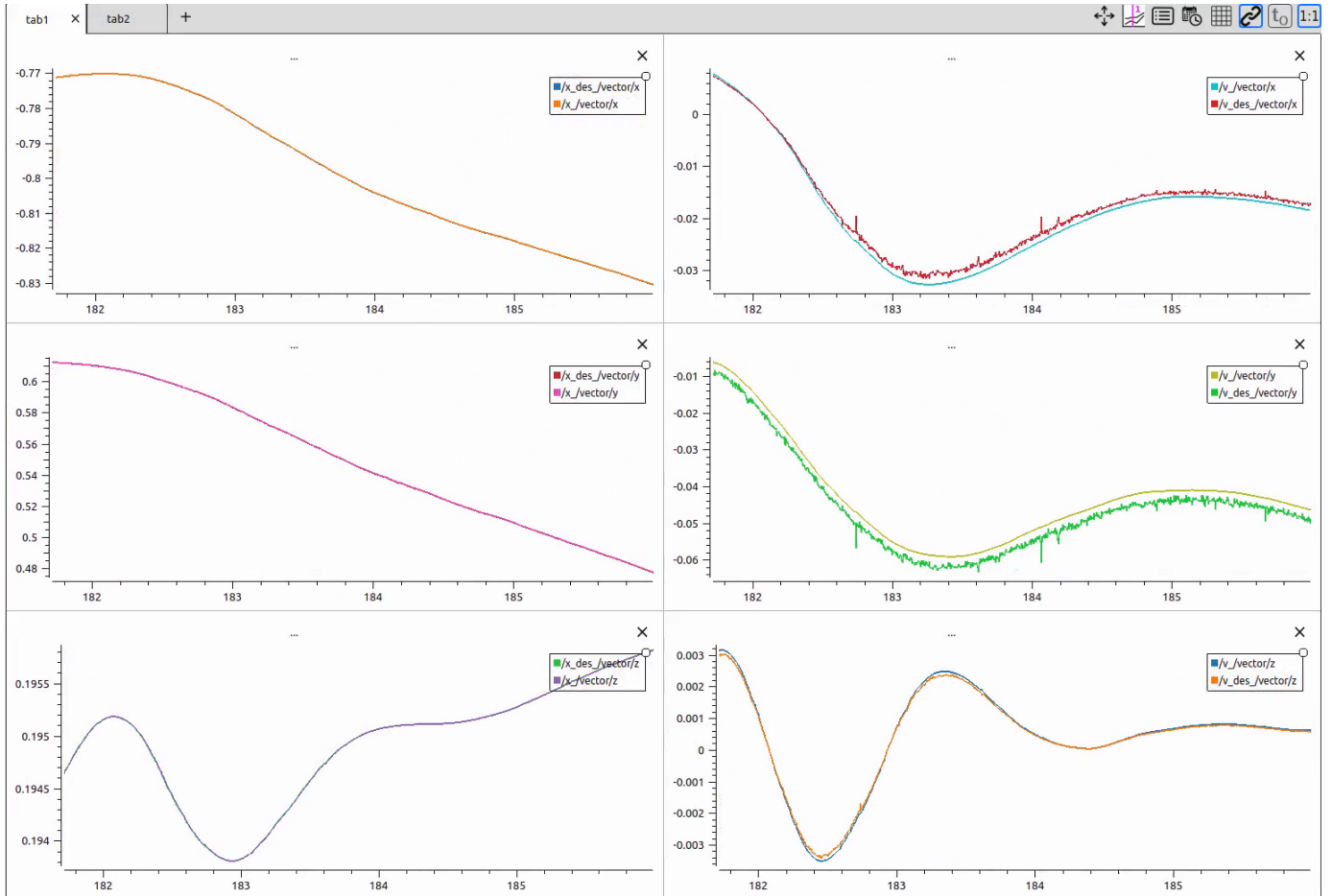


Fig. 5: Tracking of end effector position and velocity when admittance control was active